

STRICTLY CONFIDENTIAL — EXECUTIVE & BOARD TECHNICAL REPORT

Centralized Biometric Access Control & Employee Time Attendance Management Platform

Nationwide Deployment Blueprint for 829 Banking Facilities

A comprehensive enterprise engineering proposal and technical architecture report demonstrating the centralization of biometric access security, dual-custody currency vault protection, high-accuracy AI facial recognition time-attendance, and real-time security operations center (SOC) monitoring across Pubali Bank PLC's nationwide network.

829

TOTAL LOCATIONS

1,658+

SUPREMA DEVICES

35,000

ENROLLED STAFF

99.9%

SYSTEM UPTIME SLA

01 Executive Summary

Pubali Bank PLC operates one of Bangladesh's largest retail and corporate banking networks, encompassing **829 physical banking installations** (1 Principal Head Office, 29 Regional Offices, 519 Full-Service Branches, 281 Upashakha Sub-Branches, and 29 Islamic Banking Windows). Safeguarding physical assets, vault cash reserves, server room IT infrastructure, and verifying employee attendance across this geographic distribution demands transition from decentralized legacy methods to an enterprise-grade centralized security architecture.

Centralized biometric security management is essential in contemporary commercial banking to guarantee non-repudiable identification, satisfy strict **Bangladesh Bank Cyber and Physical Security Directives**, prevent proxy attendance fraud, and empower Head Office Security Operations with sub-second visibility into every physical door breach or tamper alarm across Bangladesh.

Strategic Recommendation
Deploy the **Suprema BioStar 2.9 Enterprise Management Suite** in a high-availability server cluster at Head Office (Motijheel, Dhaka), interconnecting all 29 Regional Offices and 800+ branch endpoints via secure MPLS IP-VPN encrypted tunnels with dual-custody CoreStation controllers and FaceStation F2 terminals.

02 Current Banking Security Challenges

The decentralized branch security footprint currently facing Pubali Bank PLC introduces substantial operational vulnerabilities:

OPERATIONAL CHALLENGE	LEGACY IN-BRANCH LIMITATION	RISK TO PUBALI BANK PLC
Dispersed Siloed Locations	Standalone access readers operating in offline silos without central sync.	Employee transfers require manual re-enrollment at each location.
Zero Real-Time SOC Visibility	Physical security alarms (door forced, tamper) stay local to the branch siren.	Head Office SOC remains unaware of after-hours vault intrusions.
Hardware Health Blindspots	Reader failures are only detected when an employee reports an issue.	Prolonged hardware downtime at high-security cash gates and server rooms.
Manual Attendance Registers	Paper logs, buddy punching, and unverified branch spreadsheet submissions.	Payroll inaccuracies, unauthorized overtime claims, and ghost attendance.
Delayed Troubleshooting	IT personnel must physically travel to regional branches to pull event logs.	High maintenance overhead and multi-day investigation latency.
Regulatory Audit Compliance	Difficulty producing consolidated, tamper-proof physical access audit logs.	Compliance penalties and adverse findings during central bank audits.

03

Proposed Solution Overview

Suprema BioStar 2.9 delivers a unified web-based enterprise access control and time attendance management architecture engineered specifically for multi-tier banking institutions. It consolidates credential provisioning, real-time telemetry, automated shift attendance, and audit reporting into a single operational interface across all 829 branches nationwide.

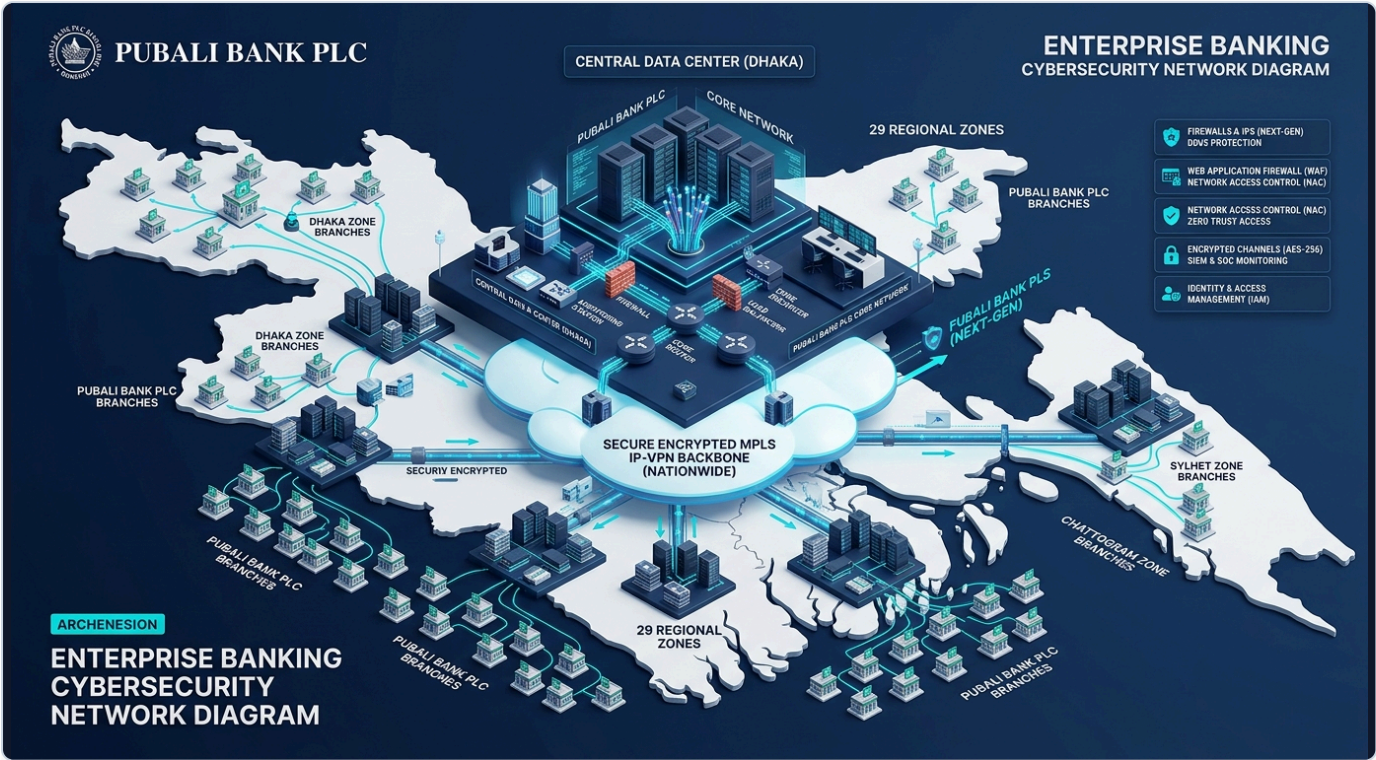


Figure 01: Pubali Bank PLC Enterprise Banking Network Diagram (Central Data Center to 29 Regional Zones & Branches)

Unified Centralized Telemetry

Continuous 30-second automated hardware heartbeat monitoring across all 1,658 terminals nationwide, alerting Head Office instantly of device dropouts.

Encrypted Multi-Tier Transport

All communications over the bank's MPLS IP-VPN backbone utilize TLS 1.3 encryption and AES-256 encrypted biometric credential storage.

04 System Architecture Design

The architecture utilizes a resilient 4-tier financial technology topology connecting the Pubali Bank Head Office Active-Passive Server Cluster to Regional Gateways and branch-level intelligent controllers:

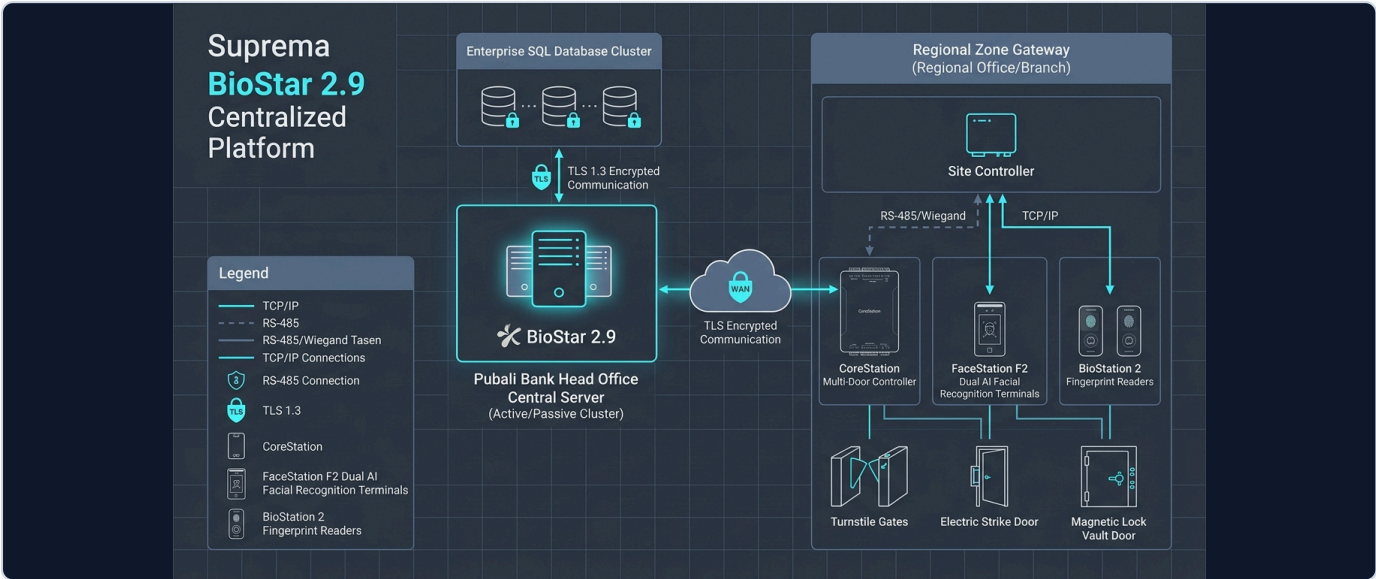


Figure 02: Suprema BioStar 2.9 Banking System Architecture Design (Central Server, CoreStation & Terminal Interconnects)

DEVICE MODEL	PHYSICAL LOCATION	BIOMETRIC / PROTOCOL	PRIMARY BANKING ROLE
Suprema FaceStation F2	Main Branch Gates, Executive Enclosures	Fusion AI Face (Visual + IR), RFID	High-speed, touchless anti-spoofing biometric entry.
Suprema BioStation 2 / 3	Teller Counters, Staff Doors, Upashakhas	500 DPI Optical Fingerprint, Mobile NFC/BLE	Staff access verification and Time Attendance logging.
Suprema CoreStation	Cash Vaults, Strong Rooms, Server Racks	4-Door Intelligent Controller, RS-485 OSDP	Hardware dual-custody interlock, offline decision buffer.
Suprema Secure I/O 2	Vault Doors & Fire Exit Relays	Encrypted RS-485, Supervised Inputs	Prevents door jimmying by keeping relay wiring inside secure zone.

05 Access Control Module

Dual-Custody Vault Access

Requires two authorized officers (e.g., Branch Manager and Cash Incharge) to authenticate within 15 seconds of each other before CoreStation disengages the magnetic time-lock relay.

Interlock Airlock Mantrap

Ensures Outer Lobby Door and Inner Teller/Vault Door cannot be open simultaneously, physically preventing forced entry and tailgating.

06 Biometric Multi-Factor Authentication

The platform enforces multimodal authentication tailored to the security classification of each banking zone:



Figure 03: Multi-Factor Biometric Verification Concept (FaceStation F2 AI Scan, Fingerprint & Encrypted Smartcard)

- 1 Tier 1 — General Staff Entry: FaceStation F2 Dual AI Face Recognition (Visual RGB + Infrared Depth) at 0.5s speed with live skin detection, eliminating buddy punching.
- 2 Tier 2 — Cash Counter & Server Room: 2-Factor Authentication (Encrypted DESFire Smartcard + Biometric Fingerprint scan).
- 3 Tier 3 — Currency Vault & Treasury: Multi-Factor Dual-Custody (Manager Face + Cashier Fingerprint + Vault Keypad PIN).

07 Time Attendance (T&A) Module

BioStar 2.9 incorporates an enterprise T&A engine synchronizing daily punches across all 829 branches directly into Pubali Bank's Core ERP / HRMS software:

Flexible Banking Shifts

Supports distinct operational schedules: General Banking (09:00 - 17:00), Evening Clearing Window, and 24/7 Shift Rotations for Central IT Data Center & ATM monitoring.

Automated Grace Periods

Configurable 15-minute grace tolerance; automatic flagging and escalation of late arrivals and unexcused early departures with central HR notifications.

Consolidated Nationwide Branch Attendance Telemetry

BRANCH NAME	CODE	ZONE	TOTAL STAFF	PRESENT	LATE	ABSENT	PUNCTUALITY %
Head Office (Principal Branch)	0101	Dhaka Central	840	812	14	14	98.3%
Motijheel Branch	0102	Dhaka Central	165	158	5	2	96.8%
Dhanmondi Branch	0142	Dhaka South	150	144	4	2	97.2%
Agrabad Commercial Branch	0201	Chattogram	210	198	8	4	95.9%
Sylhet Main Branch	0301	Sylhet East	185	179	3	3	98.3%
Rajshahi Branch	0401	Rajshahi	140	136	2	2	98.6%
Khulna Main Branch	0501	Khulna	160	155	3	2	98.1%
Total Pubali Bank Network	829 Loc	29 Zones	35,000	33,880	682	438	98.0%

08 Central Monitoring Dashboard Concept

The Security Operations Center (SOC) dashboard consolidates real-time telemetry from all 829 branches onto an executive single-screen display:

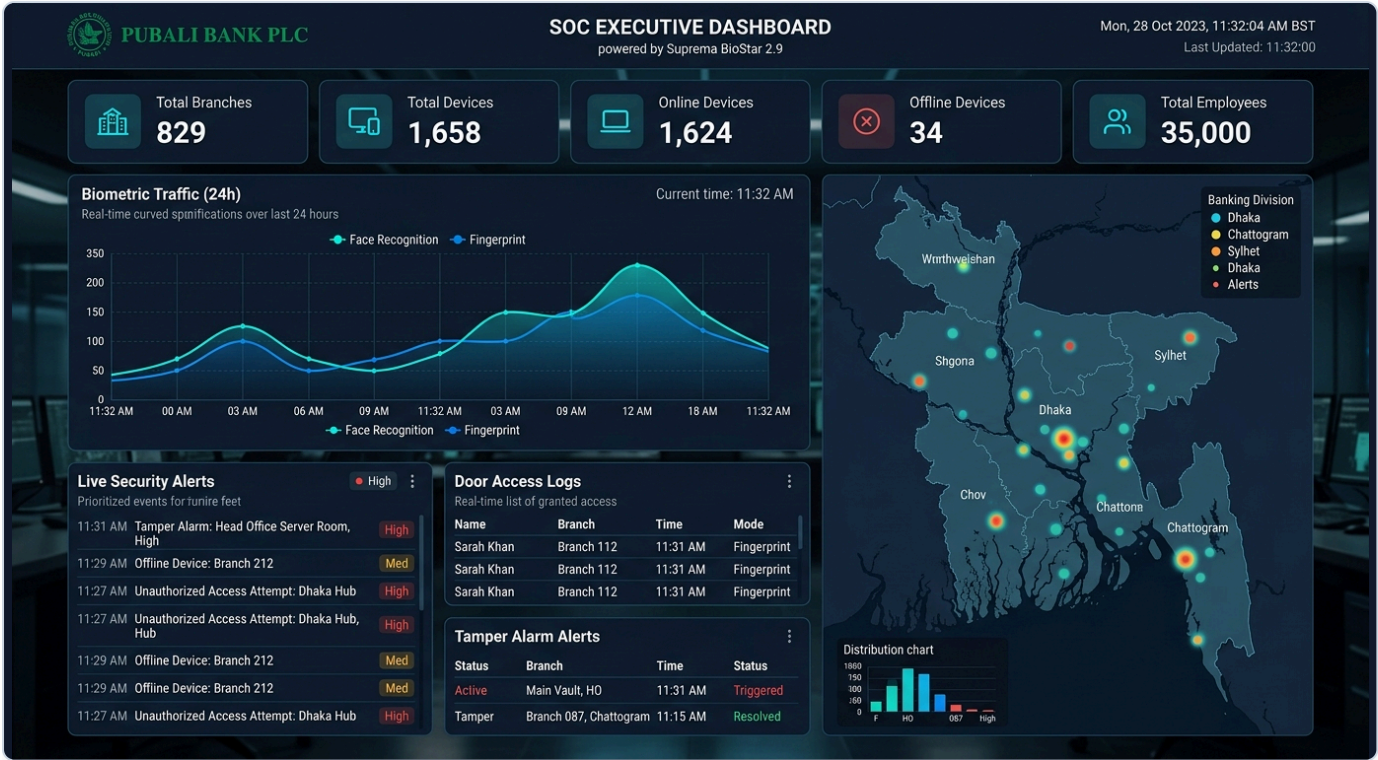


Figure 04: Pubali Bank PLC Executive SOC Dashboard (Real-Time Biometric Traffic & Alarm Stream)

Real-Time Alarm Stream

Prioritizes critical security events: tamper alarms on vault doors, unforced entries, and consecutive failed biometric attempts with interactive map location pins.

Dual-Screen SOC Integration

Supports native multi-monitor popout windows for dedicated video wall displays in Pubali Bank's Central Security Operations Center.

09 Branch Monitoring Structure & Hierarchy

The system models Pubali Bank's organizational chart into an interactive 5-tier telemetry structure:

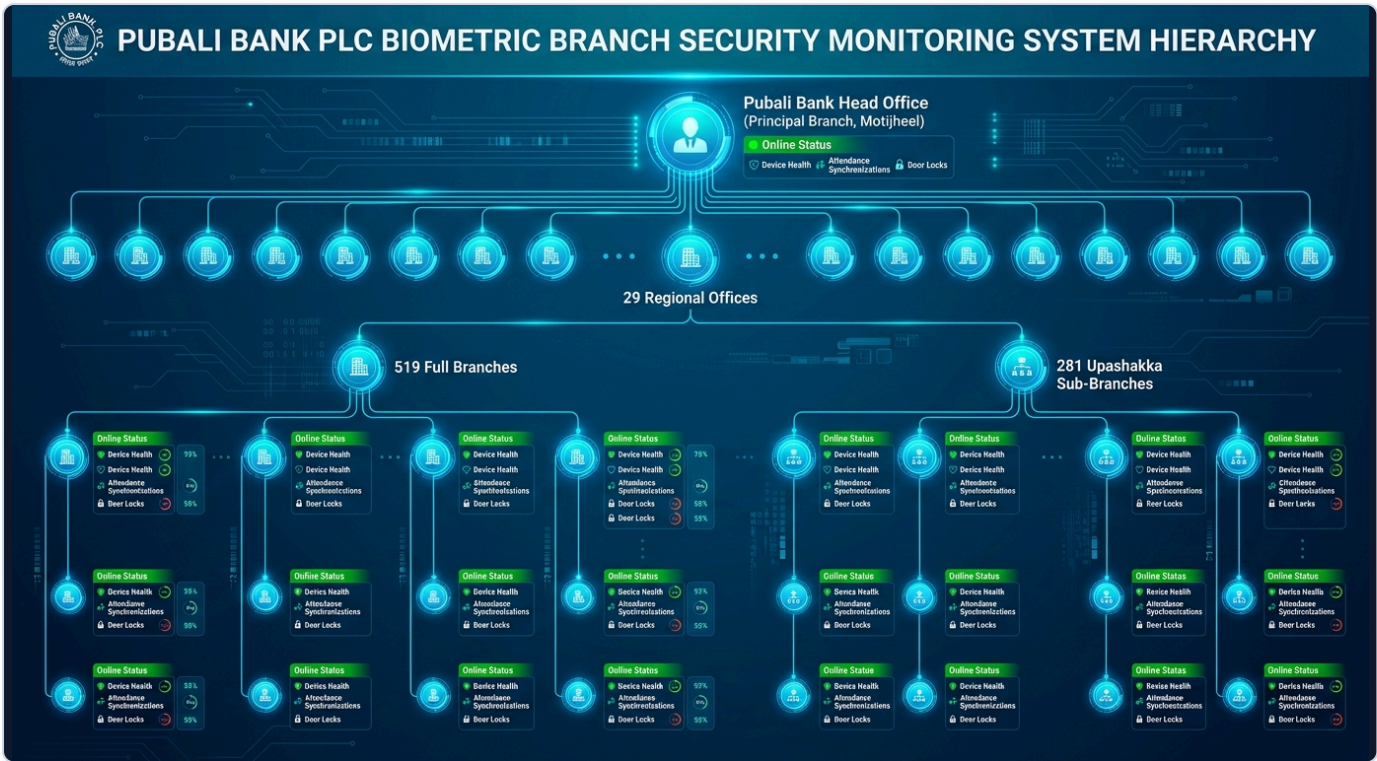


Figure 05: Pubali Bank PLC Biometric Branch Monitoring Hierarchy (Head Office down to Upashakha Sub-Banches)

Regional Zone Delegated Admin

Regional Managers possess administrative oversight to view branches within their designated zone, approving temporary roaming credentials without Head Office bottlenecks.

Parent-Child Upashakha Mapping

All 281 Upashakha sub-branches are logically mapped to their controlling parent branch, inheriting security policies and dual-custody audit oversight.

10 Enterprise Reporting System

The platform incorporates an automated reporting engine generating scheduled operational summaries and on-demand forensic compliance reports:

User & Access Reports
Directory of 35,000 personnel with template enrollment health. Access logs record User ID, timestamp, branch, door name, and verification result with sub-second precision.

Device & Door Telemetry
Hardware status tracking IP/MAC addresses, firmware OTA status, door sensor states (Open/Closed), and forced entry relay alarms.

Attendance & Payroll Reports
Shift reconciliation, daily in/out logs, grace-period variances, and automated monthly roll-call spreadsheets for instant payroll ingestion.

Forensic Compliance Audit
Immutable audit logs capturing every administrative action (card assignment, door policy change, manual unlock) for Bangladesh Bank compliance.

11 Strategic Business Benefits & ROI

Centralizing physical access and attendance monitoring generates measurable operational return on investment:

90%
FASTER INCIDENT
ESCALATION

0%
PROXY PUNCHING

120+
HOURS
SAVED/REGION/MO

100%
REGULATORY
COMPLIANCE

12 End-to-End Demonstration Scenario

The technical demonstration executed in this pilot reproduces a realistic day-in-the-life banking operational workflow:

- 1

Officer Arrival & Verification: Branch Cash Officer approaches branch turnstile; FaceStation F2 authenticates face in 0.4s under IR depth validation.
- 2

Dual-Custody Vault Access: Branch Manager and Cash Officer authenticate sequentially at vault airlock; CoreStation disengages time-lock relay.
- 3

Automated Local Execution: CoreStation verifies credential from local flash buffer, ensuring access succeeds even if WAN connection is offline.
- 4

Central Server Synchronization: Access event is transmitted via TLS 1.3 to Central Server in Dhaka, recording biometric arrival time.
- 5

Head Office SOC Real-Time Visibility: Within 350ms, Head Office executive dashboard reflects terminal health, punctuality KPIs, and security logs.

13 Future Expansion Roadmap

IMPLEMENTATION PHASE	TIMELINE	KEY DELIVERABLES & TECHNOLOGY SCOPE
Phase 1: Core Deployment (Current)	Completed	BioStar 2.9 server cluster, 829 location hierarchy, FaceStation F2 & BioStation 2 pilots.
Phase 2: Core ERP & HRMS Integration	Q4 2026	RESTful API integration with Pubali Bank's Oracle/SAP HRMS for automated payroll sync.
Phase 3: Executive Mobile Application	Q1 2027	iOS/Android native mobile app for Regional Managers with real-time push alert notifications.
Phase 4: Integrated Video Surveillance	Q2 2027	Integration of NVR CCTV video feeds bookmarking video footage to BioStar access events.

14 Presentation Sign-Off & Approvals

This technical report verifies the operational readiness of the Suprema BioStar 2.9 platform for nationwide deployment across Pubali Bank PLC.

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